

datasets, derived-view proxies, and label-aligned composites are still useful, but they carry lower claim ceilings. The full audited inventory is maintained in `docs/research/DATASET_USE_MATRIX.md`.

Table 1: Dataset-use matrix. Official, development, diagnostic, and scaffolding datasets are separated to avoid overclaiming from opened or derived-view results.

Dataset group	Pairing / modality type	Use in ELARA	Claim ceiling
MVTec 3D-AD	Real RGB + depth/XYZ	Primary naturally paired M1/Gate D benchmark; T5 strong-baseline and controlled-stress replication	Bounded in-domain/stress evidence; not external transfer or one-class SOTA
3D-ADAM	Real RGB + depth	Sealed M2 external transfer and v3 stress-regime transfer; also D13 development evidence	Opened/spent official M2; clean Gate E not confirmed, stress evidence retained
Real-IAD / Real-IAD D3	Real industrial RGB; D3 adds pseudo-3D + point cloud	D13 natural transfer attempt and D16 natural-degradation/headroom audit	D13 fails SAR; D16 all-test CW positive but stress CI crosses zero
MulSen-AD	Real RGB + infrared + point cloud	Development replication for natural positive transfer; strong future fit because modalities are complementary	Development only unless a fresh prelocked split/category is used
Eyecandies / Real3D-AD	Synthetic RGB-D-normal; 3D point-cloud benchmark	Family D failure record and earlier exploratory/mechanism evidence	Useful negative/exploratory evidence; not final transfer confirmation
VisA / MVTEC LOCO-AD	RGB with derived edge/noise proxy	Family A/C secondary diagnostics and noise-floor checks	Derived-view proxy only; cannot prove independent multimodal generalization
UNSW-NB15 / healthcare	Structured event or clinical views	Non-vision and clinical-development checks for fusion machinery	Diagnostic/development evidence; not co-observed RGB-X transfer
Fraud / behavior / NLP / vision	Label-aligned tabular/text plus small image scaffold	ELARA-Bench-LA components and utility smoke data	Mechanism/stress/scaffolding only; no natural transfer claim

2 Background and Related Work

2.1 Anomaly Detection as a Systems Problem

Deep anomaly detection has been studied across tabular, sequential, image, text, and graph data. A common difficulty is that the definition of an anomaly depends on the application, the label source, the imbalance level, and the cost of false positives and false negatives. Pang et al. [3] emphasize that anomaly detection methods cannot be evaluated without attention to context. This chapter follows that view. The purpose is not to replace domain-specific detectors. The purpose is to study what happens after those detectors produce evidence that must be fused.

2.2 Multimodal and Score-Level Fusion

Multimodal learning is often described in terms of early fusion, late fusion, hybrid fusion, and interaction-based fusion [2]. Early fusion joins features before prediction. Late fusion combines decisions or scores from separate models. Stacking trains a meta-model over domain outputs. Attention-based fusion gives the model a way to learn interactions among domains, especially when some domains are missing. The weakness is that standard attention learns a trust policy from the training distribution and does not automatically know when a domain has become unreliable at inference time.